Class-6 Chapter 1: The Wonderful World of Science

सम्पूर्ण प्रश्नों का हल NCERT नवीनतम पाठ्यक्रम पर आधारित

Class 6 Science – Chapter 1: The Wonderful World of Science

Complete Question–Answer Notes | Simple English | CBSE

01-the-wonderful-world-of-science.pdf

1. All questions and answers from the chapter

A. Think and Answer — Page 2

Q1. Name some discoveries that you find innovative. Also, mention the name of the scientist who invented it.

Answer: Some innovative discoveries and inventions are:

1. Telephone — Alexander Graham Bell.
2. Electric bulb — Thomas Alva Edison.
3. Gravity — Isaac Newton is famous for explaining the law of gravity.
4. Radioactivity — Henri Becquerel discovered natural radioactivity.
5. Penicillin — Alexander Fleming discovered it.

B. Think and Answer — Page 3

Q2. Name the dishes that are famous in your region.

Answer: Some famous dishes of Bihar are Litti-Chokha, Sattu Paratha, Dal Pitha, Khaja and Thekua.

Q3. What is the name of the device that is used to measure body temperature?

Answer: A clinical thermometer or digital thermometer is used to measure body temperature.

C. Quick Check — Page 5

State whether the following statements are True (T) or False (F).

No.	Statement	Answer
1	The paper we write on comes from wood.	True (T)
2	To observe something that sparks your interest and you want to understand is science.	True (T)
3	Scientists are those who work in a laboratory.	False (F)
4	Scientists are the people who gather information or evidence and use that information to make hypotheses.	True (T)

Explanation for Question 3: Scientists can work in laboratories, fields, forests, oceans, observatories and many other places. Anyone can also use scientific thinking in everyday life.

2. Important questions and answers from every topic

Topic 1: The World of Science

Q1. What is science?

Answer: Science is a way of thinking, observing and doing things to understand the world and discover the secrets of the universe.

Q2. Why is science important?

Answer: Science helps us understand the world, solve problems, make discoveries and develop new technologies.

Q3. How do humans begin their journey of science?

Answer: Humans begin by exploring their surroundings and asking questions about the things they see.

Q4. What does science help us understand?

Answer: Science helps us understand nature, living things, the environment and the universe.

Q5. How does science help in innovation?

Answer: Science helps us find new ideas, develop new technologies and make useful inventions.

Q6. What is meant by scientific thinking?

Answer: Scientific thinking means observing things, asking questions, finding possible answers and checking them with evidence.

Topic 2: Science Is Everywhere

Q7. Where can we find science?

Answer: We can find science everywhere, from our kitchen and playground to forests, oceans and outer space.

Q8. Give four examples that show science is everywhere.

Answer:

1. Cooking food in the kitchen.
2. Plants growing in a garden.
3. Water changing into ice.
4. Stars and planets in outer space.

Q9. Why does science have no limits to discoveries?

Answer: Every discovery gives rise to new questions and investigations. Therefore, science continues to develop and reveal new things.

Q10. Are unexpected scientific results useful? Why?

Answer: Yes, unexpected results are useful because they can change our understanding and lead to further research and discoveries.

Q11. What happens when scientists make a new discovery?

Answer: A new discovery may answer an old question and create new questions for further investigation.

Q12. How can science be found in everyday activities?

Answer: Science can be found in cooking, repairing bicycles, switching on lights, measuring temperature and many other daily activities.

Topic 3: Our Planet Earth

Q13. Which is the only planet known to support life?

Answer: Earth is the only planet currently known to support life.

Q14. Why is it important to protect Earth's environment?

Answer: It is important because Earth supports living organisms, including humans, plants and animals.

Q15. How do plants and animals survive in different environments?

Answer: Plants and animals have developed different features that help them survive in various environments and climates.

Q16. What makes Earth suitable for life?

Answer: Earth has conditions that support life, including liquid water, air and suitable temperatures.

Topic 4: Importance of Food

Q17. Why do living organisms need food?

Answer: Living organisms need food for energy, growth, development and survival.

Q18. How do plants obtain their food?

Answer: Green plants make their own food through a process called photosynthesis.

Q19. How do animals obtain food?

Answer: Animals obtain food by eating plants or other animals.

Q20. From where do human beings get food?

Answer: Human beings get food from plants and animals. Much of our food is supplied through farms.

Q21. Why should we eat a variety of foods?

Answer: We should eat a variety of foods because different foods provide different nutrients needed for proper growth and development.

Q22. Why are there different cuisines in India?

Answer: India is a diverse country. Different regions have their own food habits, ingredients, traditions and special dishes.

Q23. What is photosynthesis?

Answer: Photosynthesis is the process by which green plants use sunlight, water and carbon dioxide to make their food.

Topic 5: Things Around Us

Q24. What is temperature?

Answer: Temperature is the measure of how hot or cold something is.

Q25. What happens when water freezes?

Answer: When water freezes, it changes into ice.

Q26. What happens when water boils?

Answer: When water boils, it changes into water vapour or steam.

Q27. Why do people prefer hot water in winter and cold water in summer?

Answer: People generally prefer hot water in winter because the weather is cold, and cold water in summer because the weather is hot.

Q28. Why do we measure body temperature?

Answer: We measure body temperature to check whether a person has a fever or an unusually high or low temperature.

Q29. Name the materials from which the following objects are made.

Object	Material or source
Paper	Wood
Metal key	Metal obtained from ores
Plastic ruler	Plastic
Rubber eraser	Rubber from the rubber plant
Magnet	Metals or magnetic materials
Cotton clothes	Cotton plant
Jute clothes	Jute plant
Clay cup	Clay

Q30. What are raw materials?

Answer: Raw materials are basic materials obtained from nature or other sources and used to make useful products.

Q31. Give three examples of objects made from natural materials.

Answer:

1. Paper made from wood.
2. Cotton clothes made from cotton fibres.
3. Clay cups made from clay.

Q32. How is a plastic ruler made?

Answer: A plastic ruler is produced in a factory using plastic.

Topic 6: Scientific Method

Q33. What is the scientific method?

Answer: The scientific method is a systematic process of observing things, asking questions, predicting answers, testing them and analysing the results.

Q34. Why do we use the scientific method?

Answer: We use the scientific method to find answers to questions, solve problems and discover new information.

Q35. What are the main steps of the scientific method?

Answer: The main steps are:

1. Observe something interesting.
2. Think and ask questions about it.
3. Predict a possible answer.
4. Test the answer through experiments and observations.
5. Analyse the results to check whether the question has been answered.

Q36. What is a hypothesis?

Answer: A hypothesis is an assumption or possible explanation based on some evidence that can be tested.

Q37. Who are scientists?

Answer: Scientists are people who collect information or evidence and use it to develop and test hypotheses and build scientific theories.

Q38. Do scientists only work in laboratories?

Answer: No. Scientists can work in laboratories, forests, oceans, fields and many other places. Scientific thinking can also be used in everyday life.

Q39. How does a scientist test a hypothesis?

Answer: A scientist tests a hypothesis through experiments and observations and then analyses the results.

Q40. What is observation?

Answer: Observation is the careful examination of something to learn about it.

Q41. What is an experiment?

Answer: An experiment is a test carried out to check whether an idea or hypothesis is correct.

Q42. What is the role of evidence in science?

Answer: Evidence helps us check whether an explanation is supported by observations and experimental results.

Q43. How can we use the scientific method in daily life?

Answer: We can use it to investigate why a bulb does not glow, why a bicycle tyre becomes flat or why dal overflows while cooking.

Q44. Why is everyone a scientist in some way?

Answer: Everyone uses scientific thinking when they observe things, ask questions, look for answers and try to solve problems.

3. Additional important long-answer questions

Q45. Explain the importance of science in our daily life.

Answer: Science is an important part of our daily life. It helps us understand nature and our surroundings. We use science while cooking food, travelling, measuring temperature and using electrical appliances. Science also helps us develop medicines, machines and new technologies. It makes our lives easier and helps us solve problems.

Q46. Explain the five steps of the scientific method with an example.

Answer: The scientific method has five main steps.

1. Observation: Observe something interesting. For example, a bulb does not glow.
2. Question: Ask why the bulb is not glowing.
3. Hypothesis: Predict that the bulb may be fused or the switch may be faulty.
4. Testing: Check the switch, bulb and electrical connection safely.
5. Analysis: Study the results and identify the possible cause.

Thus, the scientific method helps us find answers in a systematic way.

Q47. Explain the importance of food for living organisms.

Answer: Food is essential for the growth and survival of living organisms. It provides energy and nutrients. Plants make their own food through photosynthesis, while animals obtain food from plants or other animals. Human beings obtain food from both plants and animals. Eating a variety of foods helps our bodies grow and develop properly.

Q48. How does science help us understand the things around us?

Answer: Science helps us understand the properties and changes of different things. For example, we learn that water changes into ice when frozen and into water vapour when heated sufficiently. We also learn about temperature, materials and how objects are made. These observations help us understand our surroundings.

Q49. Explain how scientific thinking helps solve everyday problems.

Answer: Scientific thinking helps us observe a problem, ask questions and find possible explanations. We can test these explanations through experiments or further observations. For example, if a bicycle tyre becomes flat, we can inspect it to find the source of the air leak. In this way, scientific thinking helps us solve problems logically.

Q50. Why is science a continuous process?

Answer: Science is a continuous process because every discovery can lead to new questions. Scientists carry out further observations and experiments to find answers. Even unexpected results can provide useful information. Therefore, scientific knowledge continues to grow and improve.

4. Multiple-choice questions (MCQs)

Practice quiz

0/15 answered

Choose one answer for each question. Then check your score.

1. What is science?

A. Only the study of plants
B. A way of thinking, observing and understanding
C. Only laboratory work
D. Only the study of machines

2. Which planet is known to support life?

A. Mars
B. Venus
C. Earth
D. Jupiter

3. How do green plants make their food?

A. By photosynthesis
B. By eating animals
C. By absorbing ready-made food from soil
D. By drinking milk

4. What is the measure of how hot or cold something is called?

A. Length
B. Mass
C. Temperature
D. Speed

5. What does water become when it freezes?

A. Steam
B. Ice
C. Oxygen
D. Oil

6. Which device is used to measure body temperature?

A. Barometer
B. Clinical thermometer
C. Measuring tape
D. Speedometer

7. What is a hypothesis?

A. A final result only
B. A random fact
C. A possible explanation that can be tested
D. A type of material

8. Which is the first step of the scientific method described in the chapter?

A. Analyse results
B. Conduct an experiment
C. Observe something interesting
D. Write the conclusion

9. Where can science be found?

A. Only in schools
B. Only in laboratories
C. Everywhere around us
D. Only in outer space

10. What should we do after testing a hypothesis?

A. Ignore the results
B. Analyse the results
C. Stop asking questions forever
D. Change the evidence

11. Paper is commonly made from which source?

A. Wood
B. Iron ore
C. Clay
D. Plastic

12. Which of these is a natural source of rubber?

A. Rubber plant
B. Metal ore
C. Sand
D. Coal only

13. Why do living organisms need food?

A. Only to change colour
B. For growth, energy and survival
C. Only to sleep
D. Only to move faster

14. Scientists use evidence to do what?

A. Avoid investigation
B. Test hypotheses and develop theories
C. Stop discoveries
D. Replace observations with guesses

15. What can an unexpected result lead to?

A. Further investigation
B. No learning

C. The end of all science
D. The loss of all evidence
Check answers

5. Fill in the blanks

Fill in the blanks using the correct words.

1. Science is a way of thinking and observing.
2. Earth is the only planet currently known to support life.
3. Green plants make their food through photosynthesis.
4. The measure of how hot or cold something is called temperature.
5. A hypothesis is a possible explanation that can be tested.
6. Water changes into ice when it freezes.
7. Scientists collect information and evidence.
8. The scientific method includes observation, questioning and experimentation.
9. Paper is commonly made from wood.
10. Food is necessary for growth and survival.

6. Match the following

Column A	Column B
1. Paper	a. Measures body temperature
2. Clinical thermometer	b. Made from wood
3. Photosynthesis	c. A possible explanation
4. Hypothesis	d. Plants make food
5. Ice	e. Frozen water

Answers: 1–b, 2–a, 3–d, 4–c, 5–e.

7. Very short-answer questions for revision

1. What is science? — A way of understanding the world through thinking and observation.
2. What is temperature? — A measure of how hot or cold something is.
3. What is a hypothesis? — A possible explanation that can be tested.
4. What is an experiment? — A test used to investigate an idea.
5. What is photosynthesis? — The process by which green plants make food.
6. Which planet supports life? — Earth is the only planet currently known to support life.
7. Name two materials obtained from plants. — Cotton and jute.
8. What happens when water freezes? — It changes into ice.
9. Name the first step of the scientific method. — Observation.
10. Why do we analyse results? — To find out whether the results answer our question.

8. Quick revision of the chapter

- Science helps us understand the world and the universe.
- Science is present in nature and in everyday activities.
- Earth is the only planet currently known to support life.
- Food provides energy and nutrients for growth and survival.
- Plants make food through photosynthesis.
- Different objects are made from different materials.
- Temperature tells us how hot or cold something is.
- The scientific method involves observation, questions, hypotheses, testing and analysis.
- Scientists use evidence to investigate problems and develop explanations.
- Everyone can use scientific thinking in daily life.